

Jharkhand University of Technology, Ranchi
Diploma 3rd Semester Examination, 2024 (NEP)

Subject : Computer Networks

Subject Code : CSE303

Time Allowed : 3 Hours

Full Marks : 70

Answer in your own words.

Answer any five questions in which Question No. 1 is compulsory.

The figures in the margin indicate full marks.

All questions carry equal marks.

- 1.** (i) What does TCP/IP stand for? 2×7=14
- ☒ (a) Transmission Control Protocol / Internet Protocol
 (b) Transfer Communication Process / Internet Processing
 (c) Transport Control Protocol / Internet Procedure
 (d) None of the above
- (ii) Which topology requires a multipoint connection?
- (a) Ring
 (b) ☒ Bus
 (c) Star
 (d) Mesh
- (iii) What is the primary function of a router?
- ☒ (a) To connect multiple networks and route data
 (b) To convert analog signals to digital
 (c) To store data
 (d) To amplify network signals
- (iv) What is the maximum length of an IPv6 address?
- (a) 32 bits
 (b) 64 bits
☒ (c) 128 bits
 (d) 256 bits
- (v) Which protocol is used to retrieve the email?
- (a) HTTP
 (b) FTP
 (c) SMTP
☒ (d) POP3
- (vi) The physical layer is concerned with
- ☒ (a) bit-by-bit delivery
 (b) process to process delivery
 (c) application to application delivery
 (d) port to port delivery
- (vii) ?
- (a) Router
 (b) Switch
 (c) Hub
 (d) Firewall

2. ~~(a)~~ Discuss the various causes of transmission impairment. State the transmission modes of a signal. 7+7
- ~~(b)~~ Discuss the various signals : Analog, Digital, Periodic and Non-Periodic. 7+7
3. ~~(a)~~ Discuss the various categories of Networks. 7+7
- ~~(b)~~ What is the OSI reference model? Discuss the various OSI layers and their functions. 7+7
4. ~~(a)~~ Discuss the TCP/IP reference model with a neat diagram. 7+7
- ~~(b)~~ Discuss the various TCP/IP Messages. 7+7
5. ~~(a)~~ Illustrate the various networking topologies with their merits and demerits. 7+7
- ~~(b)~~ What is NAT? Discuss how it works. 7+7
6. (a) Explain about IPV4 and compare it with IPV6. 7+7
- (b) Find the classes of each address:
- (i) 11000001 10000011 00011011 11111111
- (ii) 240.0.0.1
7. Short notes on *any four*: 3.5×4
- (a) SMTP
- (b) PoP3
- (c) Firewall
- (d) DHCP Server
- (e) VLAN
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